

Jiequan Cui

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🌐 <https://jiequancui.github.io>



Educations

- The Chinese University of Hong Kong (CUHK)
Ph.D. in Computer Science & Engineering
Shatin, Hong Kong SAR
2018 - 2022
- ShanDong University
B.E. in Computer Science
Jinnan, ShanDong
2014 - 2018
- **Ranking:** 1/141 (first 3 years)

Work Experiences

- Full Professor
Hefei University of Technology (HFUT)
Research Fellow
Nanyang Technological University (NTU)
2025.05 - Now
Hefei, China
2023.05 - 2025.04
Singapore

Research Interest

- Data-centric AI plays a crucial role in advancing towards Artificial General Intelligence (AGI). My research focuses on developing novel algorithms and theoretical foundations such as contrastive learning and generative learning to more effectively leverage data for enhancing model generalization and robustness. Besides diffusion-based generative learning, our team focuses on multi-modal agents and spatial intelligence.
- In addition, my team actively explores multi-modal alignment and robustness in large models, including LLMs, VLMs, and Agents. Our work addresses critical challenges such as jailbreak attacks, adversarial robustness, and hallucination mitigation.
- I am also broadly interested in emerging machine learning topics, including AI for science and 3D modeling, and always enthusiastic about tackling new challenges in these domains.

Awards

- IEEE TPAMI 2024 Reviewer Certificate
CVPR Outstanding Reviewer
Postgraduate studentships at CUHK
Excellent graduate in Shandong Province of China
One silver medal and one bronze medal for ACM-CCPC
One bronze medal for ACM-ICPC
National Scholarship, Ministry of Education
2025
2025
2018
2018
2016
2016
2016 & 2015

Academic Services

- Reviewer for Journals:
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).
International Journal of Computer Vision (IJCV).
IEEE Transactions on Image Processing (TIP).
IEEE Transactions on Neural Networks and Learning Systems (TNNLS).
- Reviewer for Conferences:

IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
IEEE International Conference on Computer Vision (ICCV).
European Conference on Computer Vision (ECCV).
International Conference on Learning Representations (ICLR).
Neural Information Processing Systems (NeurIPS).
International Conference on Machine Learning(ICML).

Selected Publications (Google Scholar)

* **represents equal contribution.**

1. Generalized Kullback-Leibler Divergence Loss

Jiequan Cui, Beier Zhu, Qingshan Xu, Zhuotao Tian, Xiaojuan Qi, Bei Yu, Hanwang Zhang, Richang Hong
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2026.

2. Decoupled Kullback-Leibler Divergence Loss.

Jiequan Cui, Zhuotao Tian, Zhisheng Zhong, Xiaojuan Qi, Bei Yu, Hanwang Zhang.
Conference on Neural Information Processing Systems (**NeurIPS**), 2024.

3. Generalized Parametric Contrastive Learning.

Jiequan Cui, Zhisheng Zhong, Zhuotao Tian, Shu Liu, Bei Yu, Jiaya Jia.
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023.

4. ResLT: Residual Learning for Long-tailed Recognition.

Jiequan Cui, Shu Liu, Zhuotao Tian, Zhisheng Zhong, Jiaya Jia.
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2022.

5. Parametric Contrastive Learning.

Jiequan Cui, Zhisheng Zhong, Shu Liu, Bei Yu, Jiaya Jia.
IEEE International Conference on Computer Vision (**ICCV**), 2021

6. Learnable Boundary Guided Adversarial Training.

Jiequan Cui, Shu Liu, Liwei Wang, Jiaya Jia.
IEEE International Conference on Computer Vision (**ICCV**), 2021

Full Publications (Google Scholar)

* **represents equal contribution.** † **represents corresponding author**

25. LoRA of Change: Learning to Generate LoRA for the Editing Instruction from A Single Before-After Image Pair
Xue Song, **Jiequan Cui**, Hanwang Zhang, Jiaxin Shi, Jingjing Chen, Chi Zhang, Yu-Gang Jiang
IEEE Transactions on Image Processing (**TIP**), 2026

24. Generalized Kullback-Leibler Divergence Loss

Jiequan Cui, Beier Zhu, Qingshan Xu, Zhuotao Tian, Xiaojuan Qi, Bei Yu, Hanwang Zhang, Richang Hong
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2026.

23. Rethinking VLMs for Image Forgery Detection and Localization

Shaofeng Guo, **Jiequan Cui** †, Richang Hong
Findings of the Conference on Computer Vision and Pattern Recognition (Findings of **CVPR**), 2026.

22. Reducing Class-Wise Performance Disparity via Margin Regularization

Beier Zhu, Kesen Zhao, **Jiequan Cui**, Qianru Sun, Yuan Zhou, Xun Yang, Hanwang Zhang
The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.

21. Object Fusion via Diffusion Time-step for Customized Image Editing with Single Example

Xue Song, Zhongqi Yue, **Jiequan Cui**, Hanwang Zhang, Jingjing Chen.
AAAI Conference on Artificial Intelligence (**AAAI**), 2026.

20. NeuSpring: Neural Spring Fields for Reconstruction and Simulation of Deformable Objects from Videos

Qingshan Xu, Jiao Liu, Shangshu Yu, Yuxuan Wang, Yuan Zhou, Junbao Zhou, **Jiequan Cui**, Yew-Soon Ong, Hanwang Zhang

AAAI Conference on Artificial Intelligence (**AAAI**), 2026.

19. Pushing Rendering Boundaries: Hard Gaussian Splatting.

Qingshan Xu, **Jiequan Cui** †, Xuanyu Yi, Yuxuan Wang, Yuan Zhou, Yew-Soon Ong, Hanwang Zhang
AAAI Conference on Artificial Intelligence (**AAAI**), 2026.

18. Learnable Feature Patches and Vectors for Boosting Low-light Image Enhancement without External Knowledge.

Xiaogang Xu, Jiafei Wu, Qingsen Yan, **Jiequan Cui**, Richang Hong, Bei Yu.

- IEEE International Conference on Computer Vision (**ICCV**), 2025
17. Aligned Contrastive Loss for Long-Tailed Recognition.
Jiali Ma, **Jiequan Cui**, Maeno Kazuki, Lakshmi Subramanian, Karlekar Jayashree, Sugiri Pranata, Hanwang Zhang.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR workshop**), 2025.
16. CARE Transformer: Mobile-Friendly Linear Visual Transformer via Decoupled Dual Interaction.
Yuan Zhou, Qingshan Xu, **Jiequan Cui**, Junbao Zhou, Jing Zhang, Richang Hong, Hanwang Zhang.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
15. Project-Probe-Aggregate: Efficient Fine-Tuning for Group Robustness.
Beier Zhu, **Jiequan Cui**, Hanwang Zhang, Chi Zhang.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
14. Decoupled Kullback-Leibler Divergence Loss.
Jiequan Cui, Zhuotao Tian, Zhisheng Zhong, Xiaojuan Qi, Bei Yu, Hanwang Zhang.
Conference on Neural Information Processing Systems (**NeurIPS**), 2024.
13. Robust Fine-tuning of Zero-shot Models via Variance Reduction.
Beier Zhu, **Jiequan Cui**, Hanwang Zhang.
Conference on Neural Information Processing Systems (**NeurIPS**), 2024.
12. Typicalness-Aware Learning for Failure Detection.
Yijun Liu, **Jiequan Cui**, Zhuotao Tian, Senqiao Yang, Qingdong He, Xiaoling Wang, Jingyong Su.
Conference on Neural Information Processing Systems (**NeurIPS**), 2024.
11. Instruction Tuning-free Visual Token Complement for Multimodal LLMs.
Dongsheng Wang, **Jiequan Cui**, Miaoge Li, Wang Lin, Bo Chen, Hanwang Zhang.
European Conference on Computer Vision (**ECCV**), 2024.
10. Classes Are Not Equal: An Empirical Study on Image Recognition Fairness.
Jiequan Cui, Beier Zhu, Xin Wen, Xiaojuan Qi, Bei Yu, Hanwang Zhang.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
9. Doubly Abductive Counterfactual Inference for Text-based Image Editing.
Xue Song, **Jiequan Cui**, Hanwang Zhang, Jingjing Chen, Richang Hong, Yugang Jiang.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
8. Learning Context-aware Classifier for Semantic Segmentation.
Zhuotao Tian, **Jiequan Cui**, Li Jiang, Xiaojuan Qi, Xin Lai, Yixin Chen, Shu Liu, Bei Yu, Jiaya Jia.
AAAI Conference on Artificial Intelligence (**AAAI**), 2023.
7. Generalized Parametric Contrastive Learning.
Jiequan Cui, Zhisheng Zhong, Zhuotao Tian, Shu Liu, Bei Yu, Jiaya Jia.
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023.
6. Understanding Imbalanced Semantic Segmentation Through Neural Collapse.
Zhisheng Zhong*, **Jiequan Cui***, Yibo Yang*, Xiaoyang Wu, Xiaojuan Qi, Xiangyu Zhang, Jiaya Jia.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
5. ResLT: Residual Learning for Long-tailed Recognition.
Jiequan Cui, Shu Liu, Zhuotao Tian, Zhisheng Zhong, Jiaya Jia.
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2022.
4. Parametric Contrastive Learning.
Jiequan Cui, Zhisheng Zhong, Shu Liu, Bei Yu, Jiaya Jia.
IEEE International Conference on Computer Vision (**ICCV**), 2021
3. Improving Calibration for Long-Tailed Recognition.
Zhisheng Zhong, **Jiequan Cui**, Shu Liu, Jiaya Jia.
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
2. Learnable Boundary Guided Adversarial Training.
Jiequan Cui, Shu Liu, Liwei Wang, Jiaya Jia.
IEEE International Conference on Computer Vision (**ICCV**), 2021
1. Fast and Practical Neural Architecture Search.
Jiequan Cui*, Pengguang Chen*, Ruiyu Li, Shu Liu, Xiaoyong Shen, Jiaya Jia.
IEEE International Conference on Computer Vision (**ICCV**), 2019